

Bayesian computation for statistical models with intractable normalizing constants

Yves F. Atchadé^{}, Nicolas Lartillot[†] and Christian Robert[‡]*

(First version March 2008; this version Jan. 2010)

Abstract: This paper deals with a computational aspect of the Bayesian analysis of statistical models with intractable normalizing constants. In the presence of intractable normalizing constants in the likelihood function, traditional MCMC methods cannot be applied. We propose here a general approach to sample from such posterior distributions that bypasses the computation of the normalizing constant. Our method can be thought as a Bayesian version of the MCMC-MLE approach of [8]. We illustrate our approach on examples from image segmentation and social network modeling. We study as well the asymptotic behavior of the algorithm and obtain a strong law of large numbers for empirical averages.

AMS 2000 subject classifications: Primary 60J27, 60J35, 65C40.

Keywords and phrases: Monte Carlo methods, Adaptive Markov Chain Monte Carlo, Bayesian inference, Doubly-intractable distributions, Ising model.

1. Introduction

Statistical inference for models with intractable normalizing constants poses a major computational challenge even though this problem occurs in the statistical modeling of many scientific problems. Examples include the analysis of spatial point processes ([15]), image analysis ([10]), protein design ([11]) and many others. The problem can be described as follows. Suppose we have a data set $x_0 \in (\mathcal{X}, \mathcal{B}(\mathcal{X}))$ generated from a statistical model with density

$$e^{E(x,\theta)}\lambda(dx)/Z(\theta),$$

depending on a parameter $\theta \in (\Theta, \mathcal{B}(\Theta))$, where the normalizing constant $Z(\theta) = \int_{\mathcal{X}} e^{E(x,\theta)}\lambda(dx)$ both depends on θ and is not available in closed form. Adopting a Bayesian perspective, we then

^{*}Department of Statistics, University of Michigan, email: yvesa@umich.edu

[†]Dept de Biologie, Université de Montreal, email: nicolas.lartillot@umontreal.ca

[‡]Université Paris-Dauphine and CREST, INSEE, email: xian@ceremade.dauphine.fr

take μ as the prior density of the parameter $\theta \in (\Theta, \mathcal{B}(\Theta))$. The posterior distribution of θ given x_0 is then given by

$$\pi(\theta|x_0) \propto \frac{1}{Z(\theta)} e^{E(x_0, \theta)} \mu(\theta), \quad (1)$$

where the proportionality sign means that $\pi(\theta|x_0)$ differs from the rhs by a multiplicative constant (in θ). Since $Z(\theta)$ cannot be easily evaluated, Monte Carlo simulation from this posterior distribution is particularly problematic even when using Markov Chain Monte Carlo (MCMC). [16] uses the term *doubly intractable distribution* to refer to posterior distributions of the form (1). Indeed, state-of-the-art Monte Carlo sampling methods do not allow one to deal with such models in a Bayesian framework. For example, a Metropolis-Hastings algorithm with proposal kernel Q and target distribution π , would have acceptance ratio

$$\min \left(1, \frac{e^{E(x_0, \theta')}}{e^{E(x_0, \theta)}} \frac{Z(\theta)}{Z(\theta')} \frac{\mu(\theta')}{\mu(\theta)} \frac{Q(\theta', \theta)}{Q(\theta, \theta')} \right)$$

which cannot be computed as it involves the intractable normalizing constant Z evaluated both at θ and θ' .

An early attempt dealing with this problem is the pseudo-likelihood approximation of Besag ([4]) which approximates the model $e^{E(x, \theta)}$ by a more tractable model. Pseudo-likelihood inference does provide a first approximation to the target distribution but it may perform very poorly (see e.g. [13]). Maximum likelihood inference is equally possible: for instance, MCMC-MLE, a maximum likelihood approach using MCMC, has been developed in the 90's ([7, 8]). Another related approach to find MLE estimates is Younes' algorithm ([19]) based on stochastic approximation. An interesting simulation study comparing these three methods is presented in [10], but they fail at providing more than point estimates.

Comparatively, very little work has been done to develop asymptotically exact methods for the Bayesian approach to this problem. Indeed, various approximate algorithms exist in the literature, often based on path sampling (see, e.g., [6]). Recently, [14] have shown that if exact sampling from $e^{E(x, \theta)}/Z(\theta)$ (as a density on $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$) is possible then a valid MCMC algorithm converging (1) can be constructed. See also [16] for some recent improvements. This approach relies on a clever auxiliary variable algorithm, but it requires a manageable perfect sampler and intractable normalizing constants often occur in models for which exact sampling of X is either impossible or very expensive.

In this paper, we develop a general adaptive Monte Carlo approach that samples from (1). Our algorithm generates a Θ -valued stochastic process $\{\theta_n, n \geq 0\}$ (usually not Markov) such that as $n \rightarrow \infty$, the marginal distribution of θ_n converges to (1). In principle, any method to sample from (1) needs to deal with the intractable normalizing constant $Z(\theta)$. In the auxiliary variable method of [14], computing the function $Z(\theta)$ is replaced by a perfect sampling step from $e^{E(x,\theta)}/Z(\theta)$ that somehow replaces the unknown function with an unbiased estimator. As mentioned above, this strategy works well as long as perfect sampling is feasible and inexpensive, but [13] shows that this is rarely the case for Potts models. In the present work, we follow a completely different approach building on the idea of estimating the function Z from a single Monte Carlo sampler, i.e. on the run. The starting point of the method is importance sampling. Suppose that for some $\theta^{(0)} \in \Theta$, we can sample (possibly via MCMC) from the density $e^{E(x,\theta^{(0)})}/Z(\theta^{(0)})$ in $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$. Based on such a sample, we can then estimate the ratio $Z(\theta)/Z(\theta^{(0)})$ for any $\theta \in \Theta$, for instance via importance sampling. This idea is the one behind the MCMC-MLE algorithm of [8]. However, it is well known that such estimates are typically very poor when θ gets far enough from $\theta^{(0)}$. Now, suppose that instead of a single point $\theta^{(0)}$, we use a set $\{\theta^{(i)}, i = 1, \dots, d\}$ in Θ and assume that we can sample from

$$\Lambda^*(x, i) \propto e^{E(x, \theta^{(i)})}/Z(\theta^{(i)})$$

on $\mathcal{X} \times \{1, \dots, d\}$. Then this paper shows that, in principle, efficient estimation for $Z(\theta)$ is possible for any $\theta \in \Theta$. We then propose an algorithm that generates a random process $\{(X_n, \theta_n), n \geq 0\}$ such that the marginal distribution of X_n converges to Λ^* and the marginal distribution of θ_n converges to (1).

The paper is organized as follows. In Section 2.1, we describe a general adaptive Markov Chain Monte Carlo strategy to sample from target distributions of the form (1). A full description of the practical algorithm including practical implementation details is given in Section 2.2. We illustrate the algorithm with three examples, namely the Ising model, a Bayesian image segmentation example, and a Bayesian modeling of a social network. A comparison with the method proposed in [14] is also presented. Those examples are presented in Section 4. Some theoretical aspects of the method are discussed in Section 3, while the proofs are postponed till Section 6.

2. Posterior distributions with intractable normalizing constants

2.1. A general adaptive approach

In this section, we outline a general strategy to sample from (1), which provides a unifying framework for a better understanding of the specific algorithm discussed in Section 2.2. We assume that Θ is a compact parameter space equipped with its Borel σ -algebra $\mathcal{B}(\Theta)$. Let $\mathcal{C}_+(\Theta)$ denote the set of all positive continuous functions $\zeta : \Theta \rightarrow (0, \infty)$. We equip $\mathcal{C}_+(\Theta)$ with the supremum distance and its induced Borel σ -algebra. For $\zeta \in \mathcal{C}_+(\Theta)$ such that $\exp(E(x_0, \theta)) / \zeta(\theta)$ is integrable, π_ζ denotes the density on $(\Theta, \mathcal{B}(\Theta))$ defined by

$$\pi_\zeta(\theta) \propto \frac{e^{E(x_0, \theta)}}{\zeta(\theta)}. \quad (2)$$

We assume that for those $\zeta \in \mathcal{C}_+(\Theta)$ satisfying the integrability condition, we can construct a transition kernel P_ζ with invariant distribution π_ζ and that the maps $\zeta \rightarrow P_\zeta h(\theta)$ and $\zeta \rightarrow \pi_\zeta(h)$ are measurable maps. For instance, P_ζ may be a Metropolis-Hastings kernel with invariant distribution equal to π_ζ . For a signed measure μ , we define its total variation norm as $\|\mu\|_{TV} := \sup_{|f| \leq 1} |\mu(f)|$ while, for a transition kernel P , we define its iterates as $P^0(\theta, A) = \mathbf{1}_A(\theta)$ and $P^j(\theta, A) := \int P^{j-1}(\theta, dz) P(z, A)$ ($j > 0$).

Let $\{\zeta_n, n \geq 0\}$ be a $\mathcal{C}_+(\Theta)$ -valued stochastic process (random field) defined on some probability space $(\Omega, \mathcal{F}, \Pr)$ equipped with a filtration $\{\mathcal{F}_n, n \geq 0\}$. We assume that $\{\zeta_n, n \geq 0\}$ is $\mathcal{F}_n$ -adapted. The sequence $\{\zeta_n, n \geq 0\}$ is interpreted as a sequence of estimators for Z , the true function of normalizing constants. We assume in addition that all ζ_n satisfy the integrability condition. Later on, we will see how to build such estimators in practice. At this stage, we make the following theoretical assumptions that are needed to establish our convergence results:

(A1) For any bounded measurable function $h : \Theta \rightarrow \mathbb{R}$,

$$\pi_{\zeta_n}(h) \longrightarrow \pi_Z(h) \quad a.s.$$

as $n \rightarrow \infty$, where Z is the function of normalizing constants, $Z : \theta \rightarrow \int_{\mathcal{X}} e^{E(x, \theta)} dx$.

(A2) Moreover

$$\sup_{\theta \in \Theta} \|P_{\zeta_n}(\theta, \cdot) - P_{\zeta_{n-1}}(\theta, \cdot)\|_{TV} \longrightarrow 0, \quad a.s. \text{ as } n \rightarrow \infty;$$

(A3) and there exists $\rho \in (0, 1)$ such that for all integers $n \geq 0$,

$$\sup_{k \geq 0} \sup_{\theta \in \Theta} \|P_{\zeta_k}^n(\theta, \cdot) - \pi_{\zeta_k}(\cdot)\|_{TV} \leq \rho^n.$$

Remark 2.1. The assumptions (A1-3) and the assumption that Θ is compact are strong assumptions although they hold in most of the applications that we know of when intractable normalizing constants are an issue. These assumptions can be relaxed but at the expense of greater technicalities that would overwhelm the methodological contributions of the paper.

When such a sequence of random fields is available, we can construct a Monte Carlo process $\{\theta_n, n \geq 0\}$ on $(\Omega, \mathcal{F}, \text{Pr})$ to sample from π as follows:

Algorithm 2.1. 1. Initialize $\theta_0 \in \Theta$ arbitrarily.
2. Given $\mathcal{F}_n$ and θ_n , generate θ_{n+1} from $P_{\zeta_n}(\theta_n, \cdot)$.

Then, as demonstrated in Section 6.1, this algorithm provides a converging approximation to π :

Theorem 2.1. Assume (A1-3) holds and let $\{\theta_n, n \geq 0\}$ be given by Algorithm 2.1. For any bounded measurable function $h : \Theta \rightarrow \mathbb{R}$,

$$n^{-1} \sum_{k=1}^n h(\theta_k) \longrightarrow \pi(h), \text{ a.s. as } n \rightarrow \infty.$$

We thus deduce from this generic theorem that consistent Monte Carlo sampling from π can be achieved adaptively if we manage to obtain a consistent sequence of estimates of Z . Note that the sequence $\{\zeta_n, n \geq 0\}$ of approximations does not have to be independent. We also stress that the strategy described above can actually be applied in much wider generality than the setting of intractable normalizing constants.

2.2. The initial algorithm

In this section, we specify a sequence of random field estimators $\{\zeta_n, n \geq 0\}$ that converges to Z in order to implement the principles of Algorithm 2.1.

Given the set $\{\theta^{(i)}, i = 1, \dots, d\}$ of d points in Θ , we recall that Λ^* be the probability measure on $\mathcal{X} \times \{1, \dots, d\}$ given by

$$\Lambda^*(x, i) = \frac{e^{E(x, \theta^{(i)})}}{dZ(\theta^{(i)})}, \quad (x, i) \in \mathcal{X} \times \{1, \dots, d\}. \quad (3)$$

We generalize this distribution by introducing a corresponding vector $c = (c(1), \dots, c(d)) \in \mathbb{R}^d$ of weights and defining the density function on $\mathcal{X} \times \{1, \dots, d\}$:

$$\Lambda_c(x, i) = \frac{e^{E(x, \theta^{(i)})}}{e^{c(i)} \sum_{j=1}^d Z(\theta^{(j)}) e^{-c(j)}}.$$

In the case when $c(i) = \log(Z(\theta_i))$, we recover $\Lambda_c = \Lambda^*$. We now introduce a weight function $\kappa(\theta, \theta')$ that is a positive continuous function $\Theta \times \Theta \rightarrow [0, 1]$ such that $\sum_{i=1}^d \kappa(\theta, \theta^{(i)}) = 1$ for all $\theta \in \Theta$. It will be specified below. The starting point of our algorithm is that the function Z can be decomposed as follows:

Lemma 2.1. *For $c \in \mathbb{R}^d$ and $\theta \in \Theta$, let $B(c) = \sum_{j=1}^d Z(\theta^{(j)}) e^{-c(j)}$ denote the normalizing constant of Λ_c and define the function $h_{\theta, c}^{(i)}(x, j) = e^{c(i)} e^{(E(x, \theta) - E(x, \theta^{(i)}))} \mathbf{1}_{\{i\}}(j)$. Then*

$$Z(\theta) = B(c) \sum_{i=1}^d \kappa(\theta, \theta^{(i)}) \int_{\mathcal{X} \times \{1, \dots, d\}} h_{\theta, c}^{(i)}(x, j) \Lambda_c(x, j) \lambda(dx, dj). \quad (4)$$

Proof. We have

$$\begin{aligned} Z(\theta) &= \int_{\mathcal{X}} e^{E(x, \theta)} \lambda(dx) \\ &= \sum_{i=1}^d \kappa(\theta, \theta^{(i)}) \int_{\mathcal{X}} e^{E(x, \theta) - E(x, \theta^{(i)})} e^{E(x, \theta^{(i)})} \lambda(dx) \\ &= B(c) \sum_{i=1}^d \kappa(\theta, \theta^{(i)}) \int_{\mathcal{X} \times \{1, \dots, d\}} e^{c(i)} e^{(E(x, \theta) - E(x, \theta^{(i)}))} \mathbf{1}_{\{i\}}(j) \frac{e^{E(x, \theta^{(j)})}}{e^{c(j)} B(c)} \lambda(dx, dj) \\ &= B(c) \sum_{i=1}^d \kappa(\theta, \theta^{(i)}) \int h_{\theta, c}^{(i)}(x, j) \Lambda_c(x, j) \lambda(dx, dj). \end{aligned}$$

□

The interest of the decomposition (4) is that Λ_c does not depend on θ . Therefore, if we can run a Markov chain $\{(X_n, I_n), n \geq 0\}$ with target distribution Λ_c , the quantity

$$\sum_{i=1}^d \kappa(\theta, \theta^{(i)}) \left(\frac{1}{n} \sum_{k=1}^n e^{c(i)} e^{E(X_k, \theta) - E(X_k, \theta^{(i)})} \mathbf{1}_{\{i\}}(I_k) \right)$$

will converge almost surely to $Z(\theta) B(c)^{-1}$ for any $\theta \in \Theta$ and can therefore be used in Algorithm 2.1 to sample from (1). This observation thus leads to our first practical sampler to sample from (1). To construct the Markov chain $\{(X_n, I_n), n \geq 0\}$ with target distribution Λ_c , we assume that, for each $i \in \{1, \dots, d\}$, a transition kernel $T_i^{(\mathcal{X})}$ on $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$ with invariant distribution

$e^{E(x, \theta^{(i)})}/Z(\theta^{(i)})$ is available. We use the subscript $\mathcal{X}$ to emphasize that $T_i^{(\mathcal{X})}$ is a kernel on $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$.

Algorithm 2.2. . Set $c \in \mathbb{R}^d$ arbitrarily. Let $(X_0, I_0, \theta_0) \in \mathcal{X} \times \{1, \dots, d\} \times \Theta$ be the initial state of the algorithm. Let also ζ_0 be an initial random field estimate of Z (for example $\zeta \equiv 1$).

At time $n + 1$, given (X_n, I_n, θ_n) and ζ_n ,

1. Generate X_{n+1} from $T_{I_n}^{(\mathcal{X})}(X_n, \cdot)$.
2. Generate $I_{n+1} \in \{1, \dots, d\}$ from $\Pr(I_{n+1} = i) \propto e^{E(X_{n+1}, \theta^{(i)}) - c_n(i)}$.
3. Generate θ_{n+1} from $P_{\zeta_n}(\theta_n, \cdot)$ and update ζ_{n+1} to

$$\zeta_{n+1} = \sum_{i=1}^d \kappa(\theta, \theta^{(i)}) e^{c(i)} \left(\frac{1}{n+1} \sum_{k=1}^{n+1} e^{E(X_k, \theta) - E(X_k, \theta^{(i)})} \mathbf{1}_i(I_k) \right).$$

The final step in Algorithm 2.2 thus defines a sequence of ζ_n that converges with Z . The difficulty with this approach is that in most examples, the normalizing constants $Z(\theta^{(i)})$ can vary over many orders of magnitude and, unless c is carefully chosen, the process $\{I_n, n \geq 0\}$ might never enter some of the states i . When θ is close to $\theta^{(i)}$ for one of those states, $Z(\theta)$ will be poorly estimated. This is a well-known problem in simulated tempering as well. A formal solution is to use $c = c^* = \log Z$ from the start, which is equivalent to sampling from Λ^* . But since Z is unknown, an estimate must be used instead. A solution has recently been considered in [3], building on the Wang-Landau algorithm of [18]. We follow and improve that approach here, deriving an adaptive construction of the weights c .

2.3. An adaptive version of the algorithm

In order to cover correctly the space Θ and ensure a proper approximation to $Z(\theta)$ for all θ 's, we build a scheme that adaptively adjust c such that the marginal distribution of Λ_c on $\{1, \dots, d\}$ converges to the uniform distribution. In the limiting case when this distribution is uniform, we recover $c(i) = \log Z(i)$, namely the exact normalizing constants. We use a slowly decreasing sequence of (possibly random) positive numbers $\{\gamma_n\}$ that satisfies

$$\sum_n \gamma_n = +\infty \quad \text{and} \quad \sum_n \gamma_n^2 < +\infty.$$

The following algorithm is a direct modification of Algorithm 2.2 where the weight vector c is set adaptively. It defines a non-Markovian adaptive sampler that operates on $\mathcal{X} \times \{1, \dots, d\} \times \mathbb{R}^d \times \Theta$:

Algorithm 2.3. Set $(X_0, I_0, c_0, \theta_0) \in \mathcal{X} \times \{1, \dots, d\} \times \mathbb{R}^d \times \Theta$ as the initial state of the algorithm, with ζ_0 as the initial random field estimate of Z (for example $\zeta_0 \equiv 1$).

At time $n + 1$, given $(X_n, I_n, c_n, \theta_n)$ and ζ_n :

1. Generate X_{n+1} from $T_{I_n}^{(\mathcal{X})}(X_n, \cdot)$.
2. Generate $I_{n+1} \in \{1, \dots, d\}$ from $\Pr(I_{n+1} = i) \propto e^{E(X_{n+1}, \theta^{(i)}) - c_n(i)}$.
3. Generate θ_{n+1} from $P_{\zeta_n}(\theta_n, \cdot)$
4. Update c_n to c_{n+1} as

$$c_{n+1}(i) = c_n(i) + \gamma_n \frac{e^{E(X_{n+1}, \theta^{(i)}) - c_n(i)}}{\sum_{j=1}^d e^{E(X_{n+1}, \theta^{(j)}) - c_n(j)}}, \quad i = 1, \dots, d; \quad (5)$$

and update ζ_n to ζ_{n+1} as

$$\zeta_{n+1}(\theta) = \sum_{i=1}^d \kappa(\theta, \theta^{(i)}) e^{c_{n+1}(i)} \left[\frac{\sum_{k=1}^{n+1} e^{E(X_k, \theta) - E(X_k, \theta^{(i)})} \mathbf{1}_i(I_k)}{\sum_{k=1}^{n+1} \mathbf{1}_i(I_k)} \right], \quad (6)$$

Before proceeding to the analysis of the convergence properties of this algorithm and discussing its calibration, a few remarks are in order:

Remark 2.2. 1. The last term of (6) is not defined when all the I_k 's are different from i .

In this case (which necessarily occurs in the first steps of the algorithm), it can either be replaced with zero or with an approximative Rao-Blackwellized version, namely replacing both $\mathbf{1}_i(I_k)$ terms with the probabilities $\Pr(I_k = i)$ that have been computed in Step 2. of the algorithm.

2. It is easy to see that ζ_n as given by (6) is a random field with positive sample paths and that $\{(\zeta_n, \theta_n)\}$ falls within the framework of Section 2.1. In particular, we will show below that assumption (A1-3) are satisfied and that Theorem 2.1 applies thus establishing the consistency of Algorithm 2.3.
3. Algorithm 2.3 can be seen as an MCMC-MCMC analog to the MCMC-MLE of [8]. Indeed, with $d = 1$, the decomposition (4) becomes

$$Z(\theta)/Z(\theta^{(1)}) = \mathbb{E} \left[e^{E(\theta, X) - E(X, \theta^{(1)})} \right],$$

where the expectation is taken with respect to $X \sim e^{E(x, \theta^{(1)})}/Z(\theta^{(1)})$. But, as discussed above in the introduction, $E(\theta, X) - E(X, \theta^{(1)})$ may have a large variance, and thus the resulting estimate can be terribly poor.

4. We introduce κ to serve as a smoothing factor so that the particles $\theta^{(i)}$ close to θ contribute more to the estimation of $Z(\theta)$. We expect this smoothing step to reduce the variance of the overall estimate of $Z(\theta)$. In the simulations we choose

$$\kappa(\theta, \theta^{(i)}) = \frac{e^{-\frac{1}{2h^2} \|\theta - \theta^{(i)}\|^2}}{\sum_{j=1}^d e^{-\frac{1}{2h^2} \|\theta - \theta^{(j)}\|^2}}.$$

The value of the smoothing parameter h is set by trials and errors for each example. More investigation is needed to better understand how to choose h efficiently but, as a default, it can be chosen as a bandwidth for the Gaussian non-parametric kernel associated with the sample of θ_n 's.

5. The implementation of the algorithm requires keeping track of all the samples X_k that have been generated, due to the update in (6). Since $\mathcal{X}$ can be a very high-dimensional space, we are aware of the fact that, in practice, this bookkeeping can significantly slow down the algorithm. But, in many cases, the function E takes the form $E(x, \theta) = \sum_{l=1}^K S_l(x) \theta_l$ for some real-valued functions S_l . In such cases, we only need to keep track of the sufficient statistics $\{(S_1(X_n), \dots, S_K(X_n)), n \geq 0\}$. All the examples discussed below fall within this latter category.
6. As mentioned earlier, the update of (X_n, I_n, c_n) is essentially the Wang-Landau algorithm of [3] with the following important difference. [3] propose to update c_n one component per iteration:

$$c_{n+1}(i) = c_n(i) + \gamma_n \mathbf{1}_{\{i\}}(I_{n+1}).$$

We improve on this scheme in (5) by a Rao-Blackwellization step where we integrate out I_{n+1} .

7. The proposed algorithm is not Markovian. The process $\{(X_n, I_n, c_n)\}$ becomes a nonhomogeneous Markov chain if we let $\{\gamma_n\}$ be a deterministic sequence. $\{\theta_n\}$, the main random process of interest is not a Markov chain either. Nevertheless, the marginal distribution of θ_n will typically converge to π as shown in Section 3.

We now discuss the calibration of the parameters of the algorithm.

2.4. Choosing d and the particles $\{\theta^{(i)}\}$

The algorithm 2.3 is sensitive to the choice of both d and $\{\theta^{(i)}\}$ and we provide here some guidelines. The leading factor is that the particles $\{\theta^{(i)}\}$ need to cover reasonably well the important range of the density π and be such that for any $\theta \in \Theta$, the density $e^{E(x,\theta)}/Z(\theta)$ in $\mathcal{X}$ can be well approximated by at least one of the densities $e^{E(x,\theta^{(i)})}/Z(i)$ from an importance sampling point of view. (This obviously relates to the Kullback divergence topology, see, e.g., [5]). One approach to selecting the $\theta^{(i)}$ that works well in practice consists in performing few iterations of the stochastic approximation recursion related to the maximum likelihood estimation of the model ([19]). Indeed, the normal equation of the maximum likelihood estimation of θ in the model $\exp(E(x,\theta))/Z(\theta)$ is

$$\nabla_{\theta} E(x_0, \theta) - \mathbb{E}_{\theta} [\nabla_{\theta} E(X, \theta)] = 0. \quad (7)$$

In the above, ∇_{θ} denotes the partial derivative operator with respect to θ , x_0 is the observed data set and the expectation $\mathbb{E}_{\theta}$ is with respect to $X \sim \exp(E(x,\theta))/Z(\theta)$. Younes ([19]) has proposed a stochastic approximation to solve this equation which works as follows. Given (X_n, θ_n) , generate $X_{n+1} \sim T_{\theta_n}(X_n, \cdot)$, where T_{θ} is a transition kernel on $\mathcal{X}$ with invariant distribution $\exp(E(x,\theta))/Z(\theta)$ and then set

$$\theta_{n+1} = \theta_n + \rho_n (\nabla_{\theta} E(x_0, \theta) - \nabla_{\theta} E(X_n, \theta)), \quad (8)$$

for a sequence of positive numbers ρ_n such that $\sum \rho_n = \infty$.

We thus use this algorithm to set the particles. In choosing $\{\theta^{(i)}\}$, we start by generating independently d points from the prior distribution μ . Then we carry each particle $\theta^{(i)}$ towards the important regions of π using the stochastic recursion given in (8). We typically use a constant $\rho_n \equiv \rho$, with ρ is taken around 0.1 depending on the size of Θ . A value of ρ too small will make all the particles too close together. We found few thousand iterations of the stochastic approximation to be largely sufficient.

The value of d , the number of particles, should then depend on the size and on the dimension of the compact set Θ . In particular, the distributions $e^{E(x,\theta^{(i)})}/Z(i)$ (as densities in $\mathcal{X}$) should overlap to some extent. Otherwise, the importance sampling approximation to $e^{E(x,\theta)}/Z(\theta)$ may be poor. In the simulation examples below, we choose d between 100 and 500. Note also that a

restarting provision can be made in cases when the variance of the approximation (6) is found to be too large.

2.5. Choosing the step-size $\{\gamma_n\}$

It is shown in [3] that the recursion (5) can be written as a stochastic approximation algorithm with step-size $\{\gamma_n\}$, so that in theory (see, e.g., [1]), any positive sequence $\{\gamma_n\}$ such that $\sum \gamma_n = \infty$ and $\sum \gamma_n^2 < \infty$ can be used. But the convergence of c_n to $\log Z$ is very sensitive to the choice of the sequence $\{\gamma_n\}$. For instance, if the γ_n 's are overly small, the recursive equation in (5) will make very small steps and thus converge very slowly or not at all. But if these numbers are overly large, the algorithm will have a large variance. In both cases, the convergence to the solution will be slow or even problematic. Overall, it is a well-known issue that choosing the right step-size for a stochastic approximation algorithm is a difficult problem. Here we follow [3] which has elaborated on a heuristic approach to this problem originally proposed by [18].

The main idea of this approach is that, typically, the larger γ_n is, the easier it is for the algorithm to move around the state space (as in tempering). Therefore, when starting the algorithm, γ_0 is set at a relatively large value. The sequence γ_n is kept constant until $\{I_n\}$ has visited equally well all the values in $\{1, \dots, d\}$. Let τ_1 be the first time where the occupation measure of $\{1, \dots, d\}$ by $\{I_n\}$ is approximately uniform. Then γ_{τ_1+1} is set to a smaller value (for example, $\gamma_{\tau_1+1} = \gamma_{\tau_1}/2$) and the process is iterated until γ_n become sufficiently small. At which point, we choose to switch to a deterministic sequence of the form $\gamma_n = n^{-1/2-\varepsilon}$. Combining this idea with Algorithm 2.3, we get the following straightforward implementation, where $\gamma > \varepsilon_1 > 0$, $\varepsilon_2 > 0$ are constants to be calibrated:

Algorithm 2.4. At time 0, set $(X_0, I_0, c_0, \theta_0)$ as the arbitrary initial state of the algorithm. Set $v = 0 \in \mathbb{R}^d$.

While $\gamma > \varepsilon_1$,

1. Generate $(X_{n+1}, I_{n+1}, c_{n+1}, \theta_{n+1})$ as in Algorithm 2.3.
2. For $i = 1, \dots, d$: set $v(i) = v(i) + \mathbf{1}_i(I_{n+1})$.
3. If $\max_i |v(i) - \frac{1}{d}| \leq \frac{\varepsilon_2}{d}$, then set $\gamma = \gamma/2$ and $v = 0 \in \mathbb{R}^d$.

Remark 2.3. In the application section below, we use this algorithm with the following spec-

ifications: we set the initial γ to 1, $\varepsilon_1 = 0.001$, $\varepsilon_2 = 0.2$ and the final deterministic sequence is $\gamma_n = \varepsilon_1/n^{0.7}$.

3. Convergence analysis

In this section, we characterize the asymptotics of Algorithm 2.3. We use the filtration $\mathcal{F}_n = \sigma\{(X_k, I_k, c_k, \theta_k), k \leq n\}$ and recall that Θ is a compact space equipped with its Borel σ -algebra $\mathcal{B}(\Theta)$. We also assume the following constraints:

(B1): *The prior density μ is positive and continuous and there exist $m, M \in \mathbb{R}$ such that:*

$$m \leq E(x, \theta) \leq M, \quad x \in \mathcal{X}, \theta \in \Theta. \quad (9)$$

(B2): *The sequence $\{\gamma_n\}$ is a random sequence adapted to $\{\mathcal{F}_n\}$ that satisfies $\gamma_n > 0$, $\sum \gamma_n = \infty$ and $\sum \gamma_n^2 < \infty$ Pr-a.s*

(B3): *There exist $\varepsilon > 0$, an integer n_0 , and a probability measure ν on $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$ such that for any $i \in \{1, \dots, d\}$, $x \in \mathcal{X}$, $A \in \mathcal{B}(\mathcal{X})$, $\left[T_i^{(\mathcal{X})}\right]^{n_0}(x, A) \geq \varepsilon \nu(A)$.*

Remark 3.1. In many applications, and this is the case for the examples discussed below, $\mathcal{X}$ is a finite set (that is typically very large) and Θ is a compact set. In these cases, (B1) and (B3) are easily checked. (Note that the minorization condition (B3) amounts to uniform ergodicity of the kernel $T_i^{(\mathcal{X})}$.) These assumption can be further relaxed, but the analysis of the algorithm would then require more elaborate techniques that are beyond the scope of this paper.

The following result is an easy modification of Corollary 4.2 of [3] and is concerned with the asymptotics of (X_n, I_n, c_n) . $\xrightarrow{a.s.}$ denotes convergence with probability 1.

Proposition 3.1. *Assume (B1-3). Then for any $i \in \{1, \dots, d\}$,*

$$e^{c_n(i)} \left(\sum_{k=1}^d e^{c_n(k)} \right)^{-1} \xrightarrow{a.s.} CZ(\theta^{(i)})$$

as $n \rightarrow \infty$ for some finite constant C that does not depend on i . Moreover, for any bounded measurable function $f: \mathcal{X} \times \{1, \dots, d\} \rightarrow \mathbb{R}$,

$$n^{-1} \sum_{k=1}^n f(X_k, I_k) \xrightarrow{a.s.} \Lambda^*(f)$$

as $n \rightarrow \infty$.

Under a few additional assumptions on the kernel P_ζ , the conditions of Theorem 2.1 are met.

Theorem 3.1. *Consider Algorithm 2.3 and assume (B1-3) hold. Take P_ζ as a random walk Metropolis kernel with invariant distribution π_ζ and a symmetric proposal kernel q such that there exist $\varepsilon' > 0$ and an integer $n'_0 \geq 1$ such that $q^{n'_0}(\theta, \theta') \geq \varepsilon'$ uniformly over Θ . Let $h : (\Theta, \mathcal{B}(\Theta)) \rightarrow \mathbb{R}$ be a measurable bounded function. Then*

$$n^{-1} \sum_{k=1}^n h(\theta_k) \xrightarrow{a.s.} \pi_Z(h)$$

as $n \rightarrow \infty$.

Proof. See Section 6.2. □

Remark 3.2. The uniform minorization assumption on q is only imposed here as a practical way of checking (A3). It holds on all the examples considered below due to the compactness of Θ . If P_ζ is not a random walk Metropolis kernel, that assumption should be adapted accordingly to obtain (A3).

4. Examples

4.1. Ising model

We first test our algorithm on the Ising model on a rectangular lattice, using a simulated sample x_0 . The energy function E is then given by

$$E(x) = \sum_{i=1}^m \sum_{j=1}^{n-1} x_{ij} x_{i,j+1} + \sum_{i=1}^{m-1} \sum_{j=1}^n x_{ij} x_{i+1,j}, \quad (10)$$

and $x_i \in \{1, -1\}$. In our implementation, $m = n = 64$ and we can generate the data x_0 from $e^{\theta E(x)} / Z(\theta)$ with $\theta = 0.4$ by perfect sampling through the Propp-Wilson algorithm (see, e.g., [14]). The prior used in this example is $\mu(\theta) = \mathbf{1}_{(0,3)}(\theta)$ and $d = 100$ points $\{\theta^{(i)}\}$ are generated using the stochastic approximation described in Section 2.4. As described in Section 2.5, we use the flat histogram approach in selecting $\{\gamma_n\}$ with an initial value $\gamma_0 = 1$, until γ_n becomes smaller than 0.001. Then we start feeding the adaptive chain $\{\theta_n\}$ which is run for 10,000 iterations. In updating θ_n , we use a random Walk Metropolis sampler with proposal distribution $\mathcal{U}(\theta_n - b, \theta_n + b)$ (with reflexion at the boundaries) for some $b > 0$. We adaptively update b so as to reach an

acceptance rate of 30% (see e.g. [2]). We also discard the first 1,999 simulations as a burn-in period. The results are plotted on Figure 1.a. As far as we can judge from plots (b) and (c), the sampler appears to have converged to the posterior distribution π in that the sequence is quite stable. The mixing rate of the algorithm as inferred from the autocorrelation graph in (d) seems fairly good. In addition, the algorithm yields an estimate of the partition function $\log Z(\theta)$ shown in (a) that can be re-used in other sampling problems. Note the large range of values of $Z(\theta)$ exhibited by this evaluation.

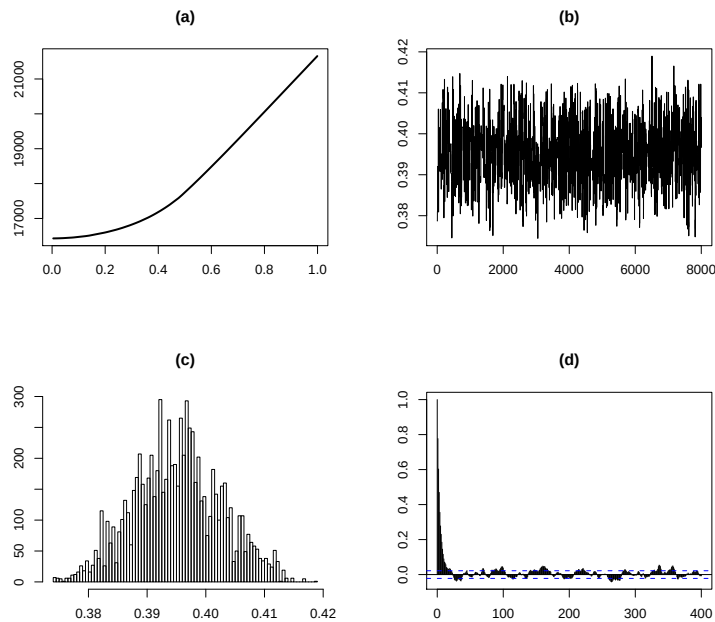


Figure 1.a: Output for the Ising model $\theta = 0.40$, $m = n = 64$. (a): estimation of $\log Z(\theta)$ up to an additive constant; (b)-(d): Trace plot, histogram and autocorrelation function of the adaptive sampler $\{\theta_n\}$.

4.2. Comparison with Moller et al. (06)

We use the Ising model described above to compare the adaptive strategy of this paper with the auxiliary variable method (AVM) of [14]. We follow the description of the AVM method given in [16]. For the comparison we use $m = n = 20$. For the adaptive strategy, we use exactly the same sampler described above. For the AVM, the proposal kernel is a Random Walk proposal from $\mathcal{U}(x - b, x + b)$ with $b = 0.05$. We have run both sampler for 10,000 iterations and reached the

following conclusions.

1. One limitation of the AVM that we have found is that the running time of the algorithm depends heavily on b and on the true value of the parameter. For larger values of b , large values of θ are more likely to be proposed and for those values, the time to coalescence in the Propp-Wilson perfect simulation algorithm can be significantly large.
2. Both samplers generate Markov chains with similar characteristics as assessed through a trace plot and an autocorrelation function. See Figure 1.b.
3. The biggest different between the two samplers is in term of computing time. For this (relatively small) example the AVM took about 16 hours to run whereas our adaptive MCMC took about 12 minutes both on an IBM T60 Laptop.

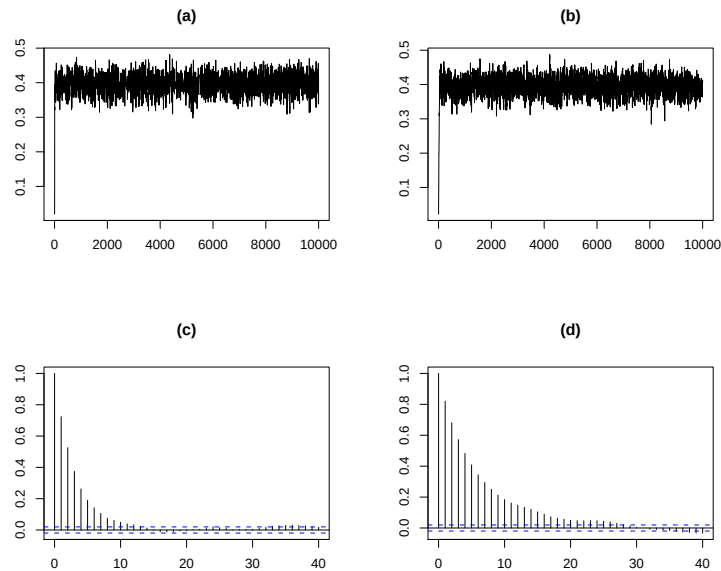


Figure 1.b: Comparison with the AVM. Trace plot and autocorrelation function. (a)-(c): the adaptive sampler. (b)-(d) The AVM algorithm. Based on 10,000 iterations.

4.3. An application to image segmentation

We use the above Ising model to illustrate an application of the methodology to image segmentation. In image segmentation, the goal is the reconstruction of images from noisy observations (see e.g. [9, 10]). We represent the image by a vector $x = \{x_i, i \in \mathcal{S}\}$ where $\mathcal{S}$ is a $m \times n$ lattice

and $x_i \in \{1, \dots, K\}$. Each $i \in \mathcal{S}$ represents a pixel, and x_i is thus the color of the pixel i , with K the number of colors. Here we assume that $K = 2$ and $x_i \in \{-1, 1\}$ is either black or white. In addition, we do not observe x directly but through a noisy approximation y . We assume here that

$$y_i|x, \sigma^2 \stackrel{\text{ind}}{\sim} \mathcal{N}(x_i, \sigma^2), \quad (11)$$

for some unknown parameter σ^2 . Even though (11) is a continuous model, it has been shown to provide a relatively good framework for image segmentation problems with multiple additive sources of noise ([10]).

As a standard assumption (see, e.g., [12]), we impose that the true image x is generated from an Ising model with interaction parameter θ . As in the previous section, θ follows a uniform prior distribution on $(0, 3)$ and we assume in addition that σ^2 has an improper prior distribution that is proportional to $1/\sigma^2 \mathbf{1}_{(0, \infty)}(\sigma^2)$. The posterior distribution (θ, σ^2, x) is then given by:

$$\pi(\theta, \sigma^2, x|y) \propto \left(\frac{1}{\sigma^2}\right)^{\frac{|\mathcal{S}|}{2}+1} \frac{e^{\theta E(x)}}{Z(\theta)} e^{-\frac{1}{2\sigma^2} \sum_{s \in \mathcal{S}} (y(s) - x(s))^2} \mathbf{1}_{(0,3)}(\theta) \mathbf{1}_{(0, \infty)}(\sigma^2),$$

where E is defined in (10).

We sample from this posterior distribution using the adaptive chain $\{(y_n, i_n, c_n, \theta_n, \sigma_n^2, x_n)\}$. The chain $\{(y_n, i_n, c_n)\}$ is updated following Steps **1.-3.** of Algorithm **2.3** and it provides the adaptive estimate of $Z(\theta)$ given by (6) (with $\{y_n, i_n\}$ replacing $\{X_n, I_n\}$). This sequence of estimates leads in its turn to update $(\theta_n, \sigma_n^2, x_n)$ using a Metropolis-within-Gibbs scheme. More specifically, given (σ_n^2, x_n) , we generate θ_{n+1} based on a random walk Metropolis step with proposal $\mathcal{U}(\theta_n - b, \theta_n + b)$ (with reflexion at the boundaries) and target proportional to $\frac{e^{\theta E(x_n)}}{\zeta_n(\theta)}$. Given θ_{n+1}, x_n , we generate σ_{n+1}^2 by sampling from the inverse Gamma distribution with parameters $(|\mathcal{S}|/2, \sum_{s \in \mathcal{S}} (y(s) - x(s))^2)/2$. At last, given $(\theta_{n+1}, \sigma_{n+1})$, we sample each $x_{n+1}(s)$ from its full conditional distribution given $\{x(u), u \neq s\}$ and $y(s)$. This conditional distribution is given by ($a \in \{-1, 1\}$)

$$p(x(s) = a|x(u), u \neq s) \propto \exp\left(\theta a \sum_{u \sim s} x(u) - \frac{1}{2\sigma^2} (y(s) - a)^2\right), \quad a \in \{-1, 1\},$$

where $u \sim v$ in the summation means that the pixels u and v are neighbors.

To test our algorithm on this model, we have generated a simulated dataset y with x generated from $e^{\theta E(x)}/Z(\theta)$ by perfect sampling. We use $m = n = 64$, $\theta = 0.40$ and $\sigma = 0.5$. For the

implementation details of the algorithm, we have made exactly the same calibration choices as in Example 4.1 above. In particular we choose $d = 100$ and generate $\{\theta^{(i)}\}$ using the stochastic approximation described in Section 2.4. The results are given in Figure 2. Once again, the sample path obtained for $\{\theta_n\}$ clearly suggests that the distribution of θ_n has converged to π with a good mixing rate, as also inferred from the autocorrelation plots.

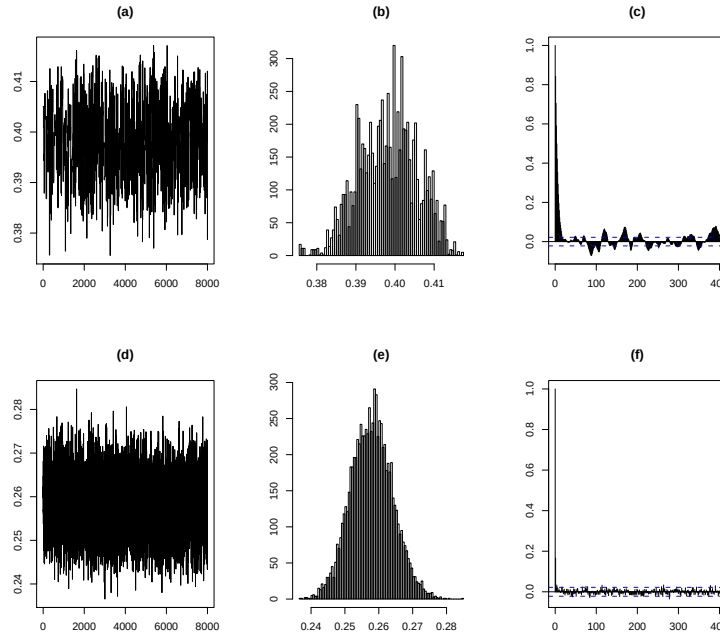


Figure 2: Output for the image segmentation model. (a)-(c): plots for $\{\theta_n\}$; (d)-(f): plots for $\{\sigma_n^2\}$.

4.4. Social network modeling

We now give an application of the method to a Bayesian analysis of social networks. Statistical modeling of social networks is a growing subject in social sciences (see e.g. [17] and the references therein for more details). The set up is the following: given n actors $I = \{1, \dots, n\}$, For each pair $(i, j) \in I \times I$, we define $y_{ij} = 1$ if actor i has ties with actor j and $y_{ij} = 0$ otherwise. In the example below, we only consider the case of a symmetric relationship where $y_{ij} = y_{ji}$ for all i, j . The ties referred to here can be of various natures. For example, we might be interested in modeling how friendships build up between co-workers or how research collaboration takes place between colleagues. Another interesting example from political science is modeling co-sponsorship

ties (for a given piece of legislation) between members of a house of representatives or parliament.

In this specific example, we study the Medici business network dataset taken from [17] which describes the business ties between 16 Florentine families. Numbering arbitrarily those families from 1 to 16, we plot the observed social network in Figure 3. The dataset contains relatively few ties between families and even fewer transitive ties.

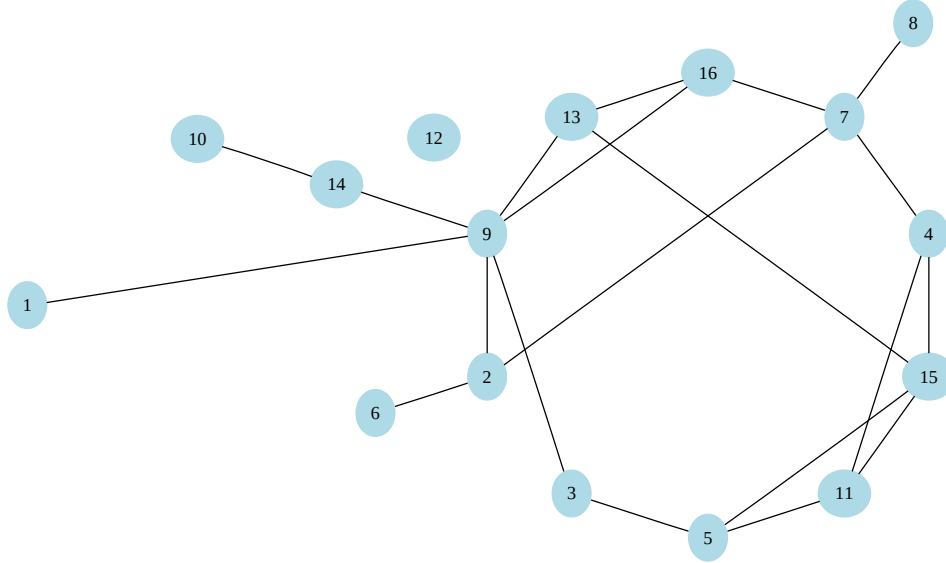


Figure 3: Business Relationships between 16 Florentine families.

One of the most popular models for social networks is the class of exponential random graph models. In these models, we assume that $\{y_{ij}\}$ is a sample generated from the parameterized distribution

$$p(y|\theta_1, \dots, \theta_K) \propto \exp\left(\sum_{i=1}^K \theta_i S_i(y)\right),$$

where $S_i(y)$ is a statistic used to capture some aspects of the network. For this example, and following [17], we consider a 4-dimensional model with statistics

$$\begin{aligned} S_1(y) &= \sum_{i < j} y_{ij}, & \text{the total number of ties,} \\ S_2(y) &= \sum_{i < j < k} y_{ik} y_{jk}, & \text{the number of two-stars,} \\ S_3(y) &= \sum_{i < j < k < l} y_{il} y_{jl} y_{kl}, & \text{the number of three-stars,} \\ S_4(y) &= \sum_{i < j < k} y_{ik} y_{jk} y_{ij}, & \text{the number of transitive ties.} \end{aligned}$$

We assume a uniform prior distribution on $D = (-50, 50)^4$ for $\theta = (\theta_1, \theta_2, \theta_3, \theta_4)$ and the corresponding posterior distribution is

$$\pi(\theta|y) \propto \frac{1}{Z(\theta)} \exp\left(\sum_{k=1}^4 \theta_k S_k(y)\right) \mathbf{1}_D(\theta). \quad (12)$$

We use Algorithm 2.3 to sample from (12). For this example, we generate 400 particles $\{\theta^{(l)}\}$ using the stochastic approximation described in Section 2.4. We use the same parameterization as in the previous examples to update (X_n, I_n, c_n) . For the adaptive chain $\{\theta_n\}$ we use a slightly different strategy, though. It turns out that some of the components of the target distribution π are strongly correlated. Therefore we sample from π in one block, using a random walk Metropolis algorithm with a Gaussian kernel $N(0, \sigma^2 \Sigma)$ (restricted to D) for $\sigma > 0$ and a positive definite matrix Σ . We adaptively set σ so as to reach the optimal acceptance rate of 30%. Ideally, we would like to choose Σ equal to Σ_π the variance-covariance of π which, of course, is not available. Instead, we adaptively estimate Σ_π during the simulation as in [2]. As before, we run (X_n, I_n, c_n) until $\gamma_n < 0.001$. Then we start $\{\theta_n\}$ and run the full chain $(X_n, I_n, c_n, \theta_n)$ for a total of 25,000 iterations. The posterior distributions of the parameters are given in Figures 4a-4d. In Table 1, we summarize those graphs by providing the sample posterior mean together with the 2.5% and 97.5% quantiles of the marginal posterior distributions. Overall, these results are consistent with the maximum likelihood estimates obtained by [17] using MCMC-MLE. The main difference appears in θ_4 which we find here to be not significant. As a by-product, the sampler gives an estimate of the variance-covariance matrix of the posterior distribution π :

$$\Sigma_\pi = \begin{bmatrix} 1.62 & -0.38 & 0.31 & -0.06 \\ -0.38 & 1.90 & -0.51 & -0.01 \\ 0.31 & -0.51 & 1.83 & -0.05 \\ -0.06 & -0.01 & -0.05 & 1.55 \end{bmatrix}.$$

Parameters	Post. mean	Post. quantiles
θ_1	-2.05	(-3.27, -0.77)
θ_2	0.96	(-0.40, 2.45)
θ_3	-1.10	(-2.78, 0.08)
θ_4	0.11	(-1.41, 1.12)

TABLE 1

Summary of the posterior distribution of the parameters. Posterior means, 2.5% and 97.5% posterior quantiles

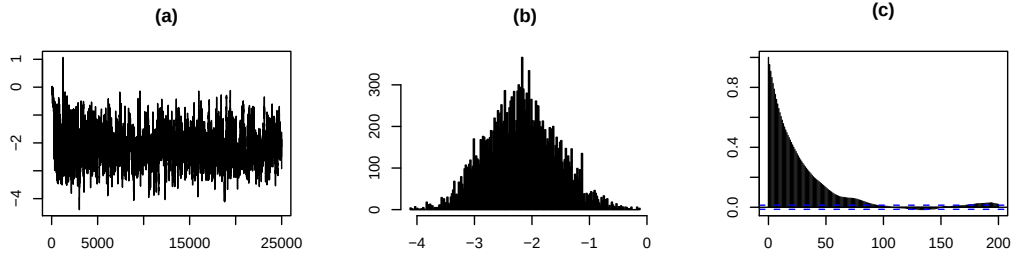


Figure 4a: The adaptive MCMC output from (12). (a)-(c): Plots for $\{\theta_1\}$. Based on 25,000 iterations.

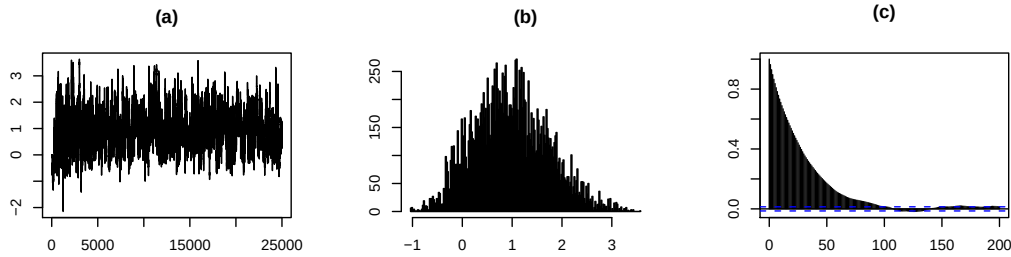


Figure 4b: The adaptive MCMC output from (12). (a)-(c): Plots for $\{\theta_2\}$. Based on 25,000 iterations.

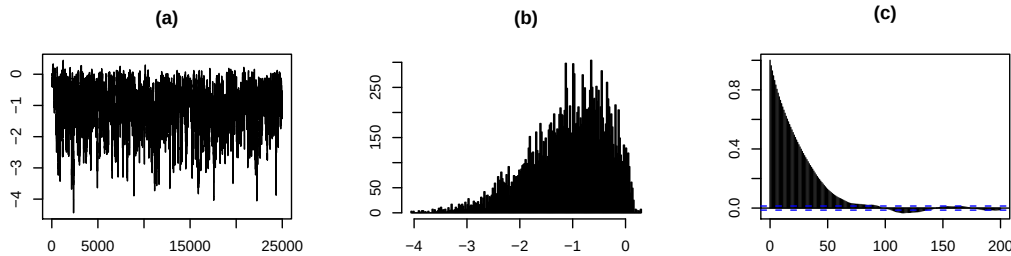


Figure 4c: The adaptive MCMC output from (12). (a)-(c): Plots for $\{\theta_3\}$. Based on 25,000 iterations.

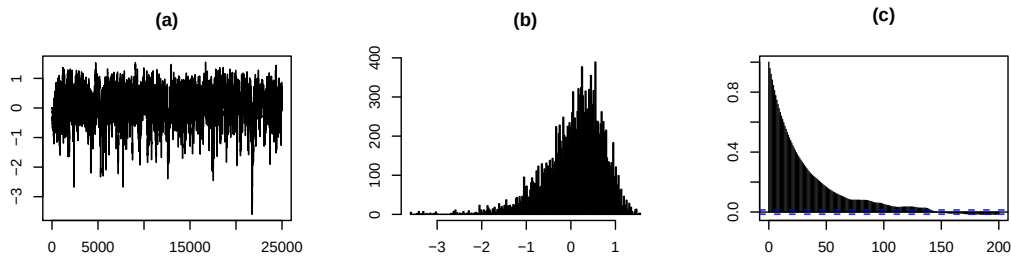


Figure 4d: The adaptive MCMC output from (12). (a)-(c): Plots for $\{\theta_4\}$. Based on 25,000 iterations.

5. Conclusion

Sampling from posterior distributions with intractable normalizing constants is a difficult computational problem. Thus far, all methods proposed in the literature but one entail approximations that do not vanish asymptotically. The only exception ([14]) based on auxiliary variables does require exact sampling in the data space, which is only possible in very specific cases settings. In this work, we have proposed an approach that both is more general than [14] and satisfies a strong law of large numbers with limiting distribution equal to the target distribution. The few applications we have presented here shows that the method is promising. It remains to be determined how the method will scale with the dimensionality and with the size of the problems, although in this respect, adaptations of the method involving annealing/tempering schemes are easy to imagine, which would allow large problems to be properly analyzed.

6. Proof

6.1. Proof of Theorem 2.1

Proof. Throughout the proof, C will denote a finite constant but whose actual value can change from one equation to the next. Also, and without any loss of generality, we will assume that θ_k is $\mathcal{F}_k$ -measurable. For $\zeta \in \mathcal{C}_+(\Theta)$, define $\bar{P}_\zeta = P_\zeta - \pi_\zeta$. Fix $h : \Theta \rightarrow \mathbb{R}$ a bounded measurable function. For $\theta \in \Theta$ and $k \geq 0$, define $g_{\zeta_k}(\theta) = \sum_{j=0}^{\infty} \bar{P}_{\zeta_k} h(\theta)$. By (A3), $g_{\zeta_k}(\theta)$ is well defined and

$|g_{\zeta_k}(\theta)| \leq (1 - \rho)^{-1} |h|_\infty$, Pr-a.s. Moreover, g_{ζ_k} satisfies the equation:

$$h(\theta) - \pi_{\zeta_k}(h) = g_{\zeta_k}(\theta) - P_{\zeta_k} g_{\zeta_k}(\theta). \quad (13)$$

For any $n, k \geq 0$, we have $\bar{P}_{\zeta_k}^n - \bar{P}_{\zeta_{k-1}}^n = \sum_{j=1}^n \bar{P}_{\zeta_k}^{n-j} (\bar{P}_{\zeta_k} - \bar{P}_{\zeta_{k-1}}) \bar{P}_{\zeta_{k-1}}^{j-1}$, Pr-a.s. We deduce from this and (A2) that $\left| \bar{P}_{\zeta_k}^n - \bar{P}_{\zeta_{k-1}}^n \right| \leq |h|_\infty \sup_{\theta \in \Theta} \left\| \bar{P}_{\zeta_k}(\theta, \cdot) - \bar{P}_{\zeta_{k-1}}(\theta, \cdot) \right\|_{TV} n \rho^{n-1}$ which implies that:

$$\left| g_{\zeta_k}(\theta) - g_{\zeta_{k-1}}(\theta) \right| \leq \frac{|h|_\infty}{(1 - \rho)^2} \sup_{\theta \in \Theta} \left\| \bar{P}_{\zeta_k}(\theta, \cdot) - \bar{P}_{\zeta_{k-1}}(\theta, \cdot) \right\|_{TV}, \quad \text{Pr-a.s.} \quad (14)$$

By the triangular inequality,

$$\left\| \bar{P}_{\zeta_k}(\theta, \cdot) - \bar{P}_{\zeta_{k-1}}(\theta, \cdot) \right\|_{TV} \leq \left\| P_{\zeta_k}(\theta, \cdot) - P_{\zeta_{k-1}}(\theta, \cdot) \right\|_{TV} + \left\| \pi_{\zeta_k}(\cdot) - \pi_{\zeta_{k-1}}(\cdot) \right\|_{TV}, \quad \text{Pr-a.s.} \quad (15)$$

Now, for any measurable function $f : \Theta \rightarrow \mathbb{R}$ such that $|f| \leq 1$ and for any $n \geq 0$,

$$\begin{aligned} \pi_{\zeta_k}(f) - \pi_{\zeta_{k-1}}(f) &= \pi_{\zeta_k} \left[P_{\zeta_k}^n (f - \pi_{\zeta_{k-1}}(f)) \right] \\ &= \pi_{\zeta_k} \left[P_{\zeta_{k-1}}^n (f - \pi_{\zeta_{k-1}}(f)) + P_{\zeta_k}^n (f - \pi_{\zeta_{k-1}}(f)) - P_{\zeta_{k-1}}^n (f - \pi_{\zeta_{k-1}}(f)) \right] \\ &= \pi_{\zeta_k} \left[P_{\zeta_{k-1}}^n (f - \pi_{\zeta_{k-1}}(f)) + \sum_{j=1}^n P_{\zeta_k}^{n-j} (P_{\zeta_k} - P_{\zeta_{k-1}}) P_{\zeta_{k-1}}^{j-1} (f - \pi_{\zeta_{k-1}}(f)) \right] \end{aligned}$$

Using (A2-3) and letting $n \rightarrow \infty$, it follows that,

$$\begin{aligned} \left\| \pi_{\zeta_k} - \pi_{\zeta_{k-1}} \right\|_{TV} &\leq 2(1 - \rho)^{-1} \sup_{\theta \in \Theta} \left\| P_{\zeta_k}(\theta, \cdot) - P_{\zeta_{k-1}}(\theta, \cdot) \right\|_{TV} \\ &\rightarrow 0, \quad \text{Pr-a.s. as } k \rightarrow \infty. \end{aligned} \quad (16)$$

Using this and (15) in (14) we can therefore conclude that:

$$\sup_{\theta \in \Theta} |g_{\zeta_k}(\theta) - g_{\zeta_{k-1}}(\theta)| \rightarrow 0, \quad \text{Pr-a.s. as } k \rightarrow \infty. \quad (17)$$

Define $S_n(h) = \sum_{k=1}^n h(\theta_k) - \pi_{\zeta_{k-1}}(h)$. Using (13), we can rewrite $S_n(h)$ as

$$\begin{aligned} S_n(h) &= \sum_{k=1}^n h(\theta_k) - \pi_{\zeta_{k-1}}(h) = \sum_{k=1}^n g_{\zeta_{k-1}}(\theta_k) - P_{\zeta_{k-1}} g_{\zeta_{k-1}}(\theta_k) \\ &= \sum_{k=1}^n g_{\zeta_{k-1}}(\theta_k) - P_{\zeta_{k-1}} g_{\zeta_{k-1}}(\theta_{k-1}) + (P_{\zeta_0} g_{\zeta_0}(\theta_0) - P_{\zeta_n} g_{\zeta_n}(\theta_n)) \\ &\quad + \left(\sum_{k=1}^n (g_{\zeta_k}(\theta_k) - g_{\zeta_{k-1}}(\theta_k)) \right) + \left(\sum_{k=1}^n (\pi_{\zeta_k}(h) - \pi_{\zeta_{k-1}}(h)) \right). \end{aligned}$$

The term $\sum_{k=1}^n g_{\zeta_{k-1}}(\theta_k) - P_{\zeta_{k-1}} g_{\zeta_{k-1}}(\theta_{k-1})$ is a $\{\mathcal{F}_n\}$ -martingale with bounded increment. From martingale limit theory, we conclude that $n^{-1} \sum_{k=1}^n g_{\zeta_{k-1}}(\theta_k) - P_{\zeta_{k-1}} g_{\zeta_{k-1}}(\theta_{k-1})$ converges to zero, Pr-a.s. as $n \rightarrow \infty$. Again, since g_{ζ_k} are uniformly bounded, $n^{-1} (P_{\zeta_0} g_{\zeta_0}(\theta_0) - P_{\zeta_n} g_{\zeta_n}(\theta_n))$ converges to zero as n goes to infinity. We conclude from (16) and (17) that the last two terms also converge to zero. In conclusion, $n^{-1} S_n(h)$ converges a.s. to zero.

Now since $n^{-1} \sum_{k=1}^n h(\theta_k) = n^{-1} S_n(h) + n^{-1} \sum_{k=1}^n \pi_{\zeta_k}(h)$, and since by (A1), $n^{-1} \sum_{k=1}^n \pi_{\zeta_k}(h) \rightarrow \pi_Z(h)$ Pr-almost surely, we conclude that the limit of $n^{-1} \sum_{k=1}^n h(\theta_k)$ as n goes to infinity is also $\pi_Z(h)$, Pr-a.s, which concludes the proof. $\square$

6.2. Proof of Theorem 3.1

Proof. We will show that (A1-3) hold and then apply Proposition 2.1. Define $\tilde{\zeta}_n(\theta) = \zeta_n(\theta) \left(\sum_{i=1}^d e^{c_n(i)} \right)^{-1}$. Using Proposition 3.1 we see that:

$$\begin{aligned} \tilde{\zeta}_n(\theta) &= \sum_{i=1}^d \kappa(\theta, \theta^{(i)}) \frac{e^{c_n(i)}}{\sum_{l=1}^d e^{c_n(l)}} \frac{\sum_{k=1}^n e^{E(X_k, \theta) - E(X_k, \theta^{(i)})} \mathbf{1}_i(I_k)}{\sum_{k=1}^n \mathbf{1}_i(I_k)} \\ &\rightarrow \sum_{i=1}^n \kappa(\theta, \theta^{(i)}) C Z(\theta^{(i)}) \frac{Z(\theta) Z(\theta^{(i)})^{-1}}{d^{-1}} \\ &= \frac{C}{d} Z(\theta), \end{aligned}$$

with probability one as $n \rightarrow \infty$. Furthermore, using (B1), we deduce easily that

$$\inf_{\theta, \theta' \in \Theta} \kappa(\theta, \theta') d^{-1} e^{m-M} \leq \sup_{\theta \in \Theta} \tilde{\zeta}_n(\theta) \leq \sup_{\theta, \theta' \in \Theta} \kappa(\theta, \theta') e^{M-m}. \quad (18)$$

It follows that for any bounded measurable function $h : \Theta \rightarrow \mathbb{R}$,

$$\pi_{\zeta_n}(h) = \pi_{\tilde{\zeta}_n}(h) = \frac{\int_{\Theta} e^{E(x_0, \theta)} \tilde{\zeta}_n^{-1}(\theta) h(\theta) d\theta}{\int_{\Theta} e^{E(x_0, \theta)} \tilde{\zeta}_n^{-1}(\theta) d\theta} \rightarrow \pi_Z(h),$$

as $n \rightarrow \infty$, by Lebesgue dominated convergence theorem. (A1) is proved.

Let $f : \Theta \rightarrow \mathbb{R}$ be a measurable function such that $|f| \leq 1$. For any $n \geq 1$, $\theta \in \Theta$, we have:

$$\begin{aligned} |P_{\tilde{\zeta}_n} f(\theta) - P_{\tilde{\zeta}_{n-1}} f(\theta)| &= \left| \int \left[\min \left(1, \frac{\tilde{\zeta}_n(\theta) e^{E(x_0, \theta') \mu(\theta')}}{\tilde{\zeta}_n(\theta') e^{E(x_0, \theta) \mu(\theta)}} \right) - \min \left(1, \frac{\tilde{\zeta}_{n-1}(\theta) e^{E(x_0, \theta') \mu(\theta')}}{\tilde{\zeta}_{n-1}(\theta') e^{E(x_0, \theta) \mu(\theta)}} \right) \right] \times \right. \\ &\quad \left. f(\theta') q(\theta, \theta') d\theta' \right| \\ &\leq \int \left| \frac{\tilde{\zeta}_n(\theta)}{\tilde{\zeta}_n(\theta')} - \frac{\tilde{\zeta}_{n-1}(\theta)}{\tilde{\zeta}_{n-1}(\theta')} \right| \frac{e^{E(x_0, \theta') \mu(\theta')}}{e^{E(x_0, \theta) \mu(\theta)}} q(\theta, \theta') d\theta' \\ &\leq C \sup_{\theta, \theta' \in \Theta} |\tilde{\zeta}_n(\theta) - \tilde{\zeta}_{n-1}(\theta)|, \end{aligned}$$

for some finite constant C . For the first inequality we use $|f| \leq 1$ and the fact that for any $a, x, y \geq 0$, $|\min(1, ax) - \min(1, ay)| \leq a|x - y|$; whereas in the last inequality, we use (B1) and (18).

We now bound the term $\sup_{\theta, \theta' \in \Theta} |\tilde{\zeta}_n(\theta) - \tilde{\zeta}_{n-1}(\theta)|$. Let $w_n(i) = e^{c_n(i)} \left(\sum_{l=1}^d e^{c_n(l)} \right)^{-1}$, $v_n(i) = \sum_{k=1}^n e^{E(X_k, \theta) - E(X_k, \theta^{(i)})} \mathbf{1}_i(I_k) (\sum_{k=1}^n \mathbf{1}_i(I_k))^{-1}$, so that $\tilde{\zeta}_n(\theta) = \sum_{i=1}^d \kappa(\theta, \theta^{(i)}) w_n(i) v_n(i)$. We have:

$$|\tilde{\zeta}_n(\theta) - \tilde{\zeta}_{n-1}(\theta)| \leq \left| \sum_{i=1}^d \kappa(\theta, \theta^{(i)}) (w_n(i) - w_{n-1}(i)) v_n(i) \right| + \left| \sum_{i=1}^d \kappa(\theta, \theta^{(i)}) w_{n-1}(i) (v_n(i) - v_{n-1}(i)) \right|.$$

The terms $w_n(i)$ satisfies $|w_n(i) - w_{n-1}(i)| \leq 2e^{\gamma_0} \gamma_n$ whereas the term $v_n(i)$ satisfies $|v_n(i) - v_{n-1}(i)| \leq e^{M-m} (\sum_{l=1}^n \mathbf{1}_i(I_l))^{-1}$. It follows that

$$\sup_{\theta, \theta' \in \Theta} |\tilde{\zeta}_n(\theta) - \tilde{\zeta}_{n-1}(\theta)| \leq C \left(\gamma_n + \frac{1}{\sum_{l=1}^n \mathbf{1}_i(I_l)} \right).$$

(A2) follows.

For $n \geq 1$, $\theta \in \Theta$ and $A \in \mathcal{B}(\Theta)$,

$$\begin{aligned} P_{\zeta_n}(\theta, A) &\geq \int_A \min \left(1, \frac{\zeta_n(\theta) e^{E(x_0, \theta')} \mu(\theta')}{\zeta_n(\theta') e^{E(x_0, \theta)} \mu(\theta)} \right) q(\theta, \theta') d\theta' \\ &\geq \left[\inf_{\theta, \theta' \in \Theta} \frac{\zeta_n(\theta) e^{E(x_0, \theta')} \mu(\theta')}{\zeta_n(\theta') e^{E(x_0, \theta)} \mu(\theta)} \right] q(\theta, A) \geq \delta q(\theta, A), \end{aligned}$$

for some $\delta > 0$, using (B1) and (18). The constant δ does not depend on n nor θ . Therefore, if $q^{n_0}(\theta, \theta') \geq \varepsilon$ then for any $n \geq 1$, $\theta \in \Theta$, $\|P_{\zeta_n}^j(\theta, \cdot) - \pi_{\zeta_n}(\cdot)\|_{TV} \leq (1 - \delta)^{j/n_0}$ which implies (A3). $\square$

Acknowledgments: We are grateful to the referees and to Mahlet Tadesse for very helpful comments that have improved the quality of this work.

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